

Project Charter

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Project Title	Universal Automotive Management App
Business Objective	The objective of this project is to develop an application that helps vehicle owners better manage and understand the maintenance of their vehicles. During market research interviews, many individuals identified issues in forgetting or delaying routine maintenance, having difficulty knowing when maintenance should be performed, not keeping organized records of previous repairs, and having difficulty tracking how much money they have spent maintaining their vehicles. The app will provide the users with a centralized place to track their vehicle maintenance, receive reminders for upcoming services, record maintenance and repair expenses, and access information about vehicle parts and maintenance needs specific to their vehicle.
Scope Boundary	In Scope: • User account creation and login. • Ability to add and manage one or more vehicles. • Storage of basic vehicle information such as make, model, year, and mileage. • Recommended maintenance based on vehicle information and mileage. • Application keeps track of miles user travelled using the location of the mobile device. • Maintenance reminders for routine services such as oil changes, tire rotations, brakes, etc. • Maintenance and repair history for each vehicle. • Recording maintenance expenses, including service performed, date, mileage, and cost. • A summary of maintenance costs for each vehicle. • Information to help users identify commonly needed vehicle parts and maintenance services. • An AI-based assistant that can answer basic questions related to vehicle maintenance and parts. Out of Scope: • Direct vehicle diagnostics through the vehicle's onboard computer. • Automatic repair or maintenance of vehicles. • Purchasing or selling vehicles. • Direct purchasing of vehicle parts through the application. • Guaranteeing that AI-generated maintenance or repair recommendations are equal to a professional mechanic's diagnosis. Deferred: • Scheduling or processing payments with repair shops. • Shared vehicle access among users • Resale/trade-in estimate for the vehicle
Constraints	• The project must be completed within the timeframe established for the course.

	• Development will be limited by the size, experience, and availability of the student development team. • Vehicle maintenance information may vary depending on make, model, year, mileage, driving conditions, and manufacturer recommendations. • Access to detailed vehicle and parts databases may be limited • AI-generated responses must be limited to informational assistance and should not be presented as guaranteed mechanical diagnoses.
Assumptions	• Users will provide accurate vehicle information, including make, model, year, and current mileage. • Users will manually record completed maintenance and repair information when necessary. • Users will allow their location to be shared with the application. • Maintenance information will be available for the vehicle types supported by the application. • Users will have access to an internet-connected computer or mobile device. • Most users will benefit from having maintenance schedules, history, expenses, and vehicle information organized in one location. • The application will initially focus on common passenger vehicles rather than specialized, commercial, or heavy-duty vehicles.
Success Criteria	• Users can successfully create and manage their vehicles within the application. • Users can view upcoming recommended maintenance for their vehicles. • Users can record completed maintenance and repair services. • Users can view the maintenance history of each vehicle. • Users can record and review maintenance and repair expenses. • The system successfully provides reminders for upcoming maintenance. • Users can obtain useful basic maintenance and parts information through the application. • The major features defined for the project's initial release are implemented and function correctly by the project deadline. • User testing shows that users can understand and use the application's core features without significant difficulty. • Final user feedback indicates that the application provides a more organized way to manage vehicle maintenance than the methods users previously relied on.
PM Authority	• Assign and coordinate project tasks among team members. • Establish and maintain the project schedule, milestones, and deadlines. • Monitor project progress and identify schedule, scope, and resource issues. • Organize team meetings and project status updates. • Approve minor changes that do not significantly affect the project's scope. • Escalate major scope changes, delays, or other significant issues to the project team and instructor for approval when necessary.

	• Ensure project documentation and deliverables are completed and submitted on time. • Facilitate the resolution of disagreements or project issues among team members. • Coordinate communication between team members and other project stakeholders. Major changes to the project's scope, objectives, or deliverables require agreement from the project team and, when applicable, approval from the instructor.
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Stakeholder Register		
<i>Name</i>	<i>Title</i>	<i>Role / Responsibility</i>
Dr. Mejia	Instructor	Approves scope changes, grades deliverables
Andres Romero	Project Team Member	Responsible for project deliverables
Randal Guzman	Project Team Member	Responsible for project deliverables
Esteban Navar	Project Team Member	Responsible for project deliverables
Carlos Adrian Yanez Duron	Project Team Member	Responsible for project deliverables
Vehicle Owners	End User	Uses the application itself